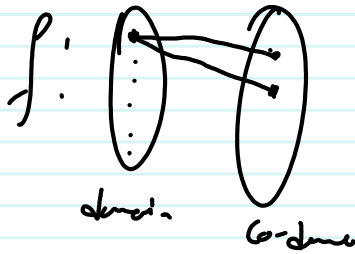
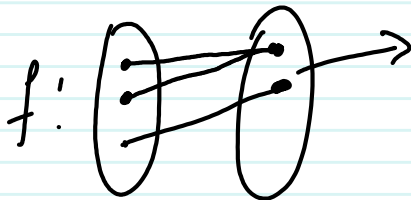


Math I

open session



one-one - f_n



$f: A \rightarrow B$ when $a, b \in A$ and

$$f(a) = f(b) \Rightarrow \underline{a = b}$$

$$a, b \in A, \quad \underline{a \neq b} \Rightarrow \underline{f(a) \neq f(b)}$$

$$f(x) = x^3, \quad 2x^3, \quad \underline{2x+5}$$

$$f: \mathbb{R} \rightarrow \mathbb{R}$$

$$\underline{a, b \in \mathbb{R}} \text{ such}$$

$$f(a) = f(b) \\ 2a+5 = 2b+5$$

function

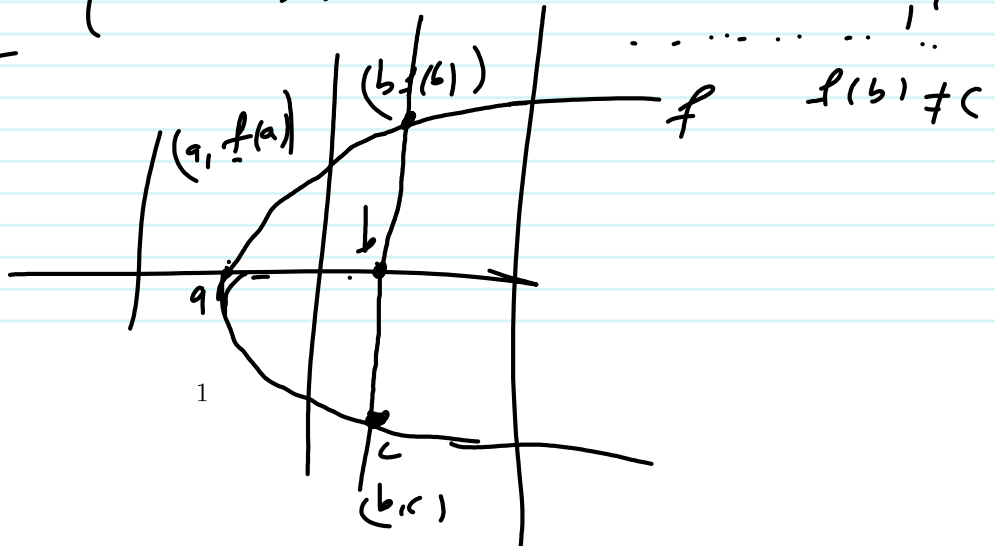
$$\Rightarrow \underline{\underline{a = b}}$$

Vertical Test

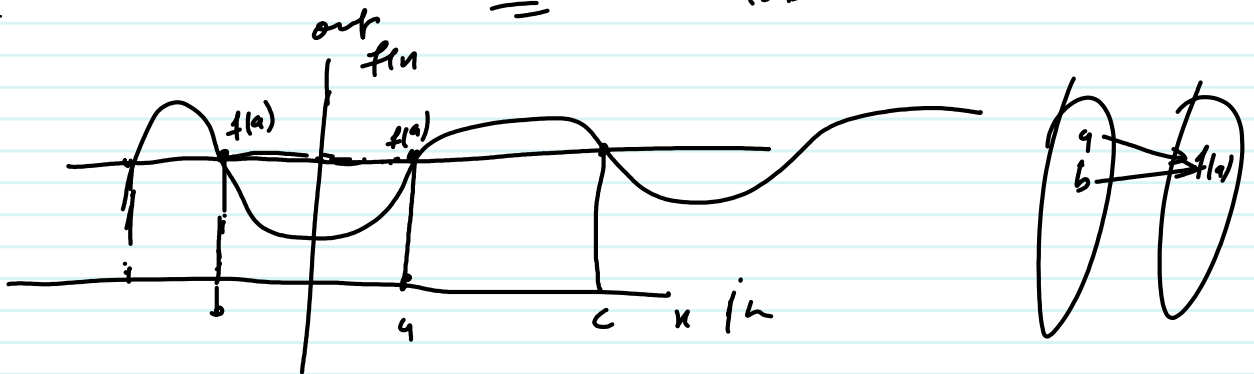
To check a relation is f_n or not

A relation $f: A \rightarrow B$ $f = \{(a, f(a)), (b, f(b)), (b, c)\}$

Vertical
line Test



Horizontal line test. To check a function
one-one or not



$$f(x) = \frac{4x+3}{3x-5}$$

$a, b \in \text{Domain} \mid \mathbb{R} \setminus \{5/3\}$

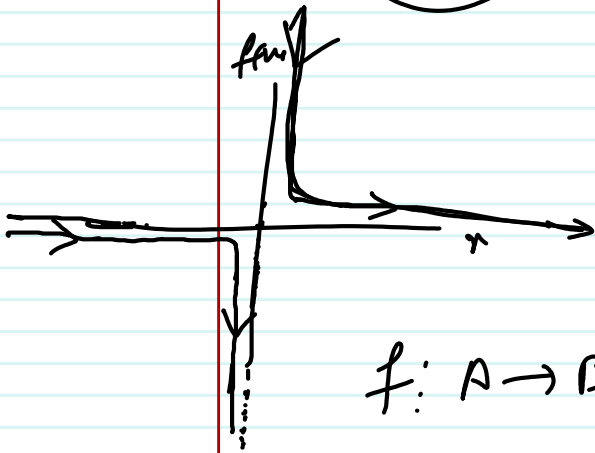
or $f(a) = f(b)$

$$\Rightarrow \frac{4a+3}{3a-5} = \frac{4b+3}{3b-5}$$

$$\Rightarrow \cancel{12a^2} - \cancel{20a} + 4b - \cancel{15} = \cancel{12ab} + \underline{4a} - \cancel{20b} - \cancel{15}$$

$$2ab = 2a$$

$$\underline{b=a}$$



$f: A \rightarrow B$, increasing / decreasing

one-one..

$$\frac{a > b}{a \neq b} \quad \underline{a, b \in A} \quad \underline{f(a) > f(b)}$$

increasing

$$\Rightarrow \underline{f(a) \neq f(b)}$$

one-one.

$$\frac{a > b}{a \neq b} \quad \rightarrow \quad f(a) < f(b)$$

decreasing.

$$\Rightarrow f(a) \neq f(b)$$

one-one..

$$a > b \quad \forall a, b \in A \quad \begin{matrix} \downarrow & \downarrow \\ f(a) & f(b) \end{matrix} \quad \text{increasing}$$

$$a > b \quad \begin{matrix} \downarrow & \downarrow \\ f(a) & f(b) \end{matrix} \quad \text{decreasing}$$

$$f(x) = 5x$$

$$x=2, 5$$

$$\frac{5 > 2}{a > b}$$

$$f(5) = 25$$

$$f(2) = 10$$

$$f(5) > f(2)$$

$$25 > 10$$

increasing..

$2n$
 n^2
 $\ln 2.5 \dots$
 $\ln 2.5$
 $\sim$

$f(x) = -5x$

$e = 2.7 \dots$
 $2 < e < 3$

$f(x)$

$f(x) = a^x$

$x \in \mathbb{R}$

$(x^2) \Rightarrow (2^x)$
 $(2^x) \Rightarrow (2^x)$

exponentiation

a^x

$0 < a < 1$
 $a > 1$

$a \neq 0$
 $a \neq 1$
 $a > 0$
 $a = -2$

$(-2)^{1/2} =$

$\frac{1}{3} = \frac{1}{3}$

$f(x) = a^x$

$0 < a < 1$
 $a > 1$

$(-2)^x, x \in \mathbb{R}$

$3^2 = 9 \downarrow$
 $3^3 = 27 \downarrow$
 $3^4 = 81 \downarrow$

$f(x) = 3^x, x \in \mathbb{R}$

$(-2)^{1/2}$
 $(\sqrt{-2})$



$a > 1$
 $x \rightarrow \infty$
 $x \rightarrow -\infty$

$0 < a < 1$

$a = 0.5 = \frac{1}{2}$

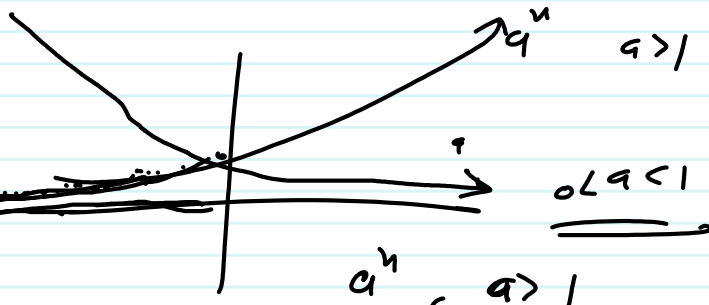
$(\frac{1}{2})^5 = \frac{1}{32}$
 $(\frac{1}{2})^2 = \frac{1}{4}$

$(\frac{1}{2})^{-2} =$

$2^2 = 4$

$(\frac{1}{2})^x$

$3^x = 0$
 $\Rightarrow$



$(9/4)^x$

$a > 1$
 $a > 1$

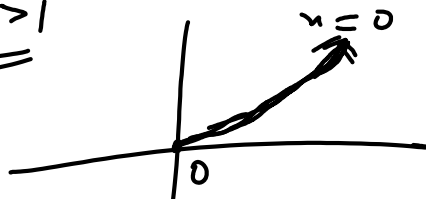
$(1/4)^x > 1$

$$f(n) = n^{9/4} \quad \underline{9/4 > 1}$$

$$n = -1 \rightarrow \infty$$

$$(-1)^{9/4} = (-1)^{9 \cdot 1/4} = (-1)^{9/4}$$

$$4^t \times 6^s$$



$$\underline{n > 6}$$

- (I) $a^t \times a^s = a^{t+s}$
- (II) $(a^t)^s = a^{ts}$
- (III) $a^t / a^s = a^{t-s}$
- (IV) $a^s b^s = (ab)^s$

$$\underline{a > 1, 0 < a < 1}$$

$$f(n) = \left(\frac{1}{7^n} + 15 \right) \rightarrow \underline{15}$$

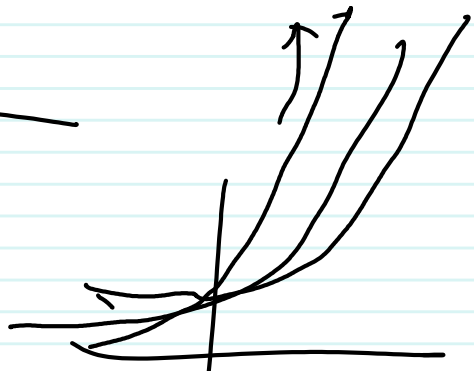
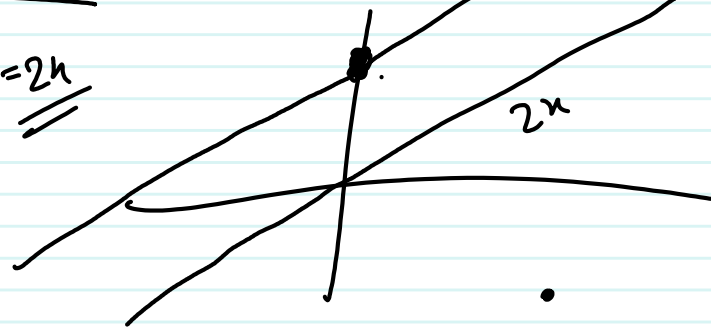
$$a=2, b=4$$

$$2^5 \times 4^{10} =$$

$$2^5 \times (2^2)^{20} =$$

$$f(n) = a^n$$

$$y(x) = 2x$$



$$(2^2 \times 3^3)^5 = (2^2)^5 \times (3^3)^5 \quad \begin{matrix} a=2 & 4 \\ a=3 & 9 \\ a=5 & 25 \end{matrix}$$

$$= 2^{10} \times 3^{15}$$

$$\underline{n_1 \neq n_2 \Rightarrow f(n_1) \neq f(n_2)}$$

$$\underline{f(n) \neq f(n_1) \Rightarrow n_1 \neq n - x}$$

$$a \Rightarrow \underline{e > 1}$$

$$2 < e < 3$$

$$2.7 \dots$$

$$\underline{a > 1}$$

$$n^2$$

$$2^{10} \times 3^{15} \times (4^7) \times 5^3$$

$$2^{14} \times 5^3$$

$$2^4 \times 3^{15} \times 5^3$$

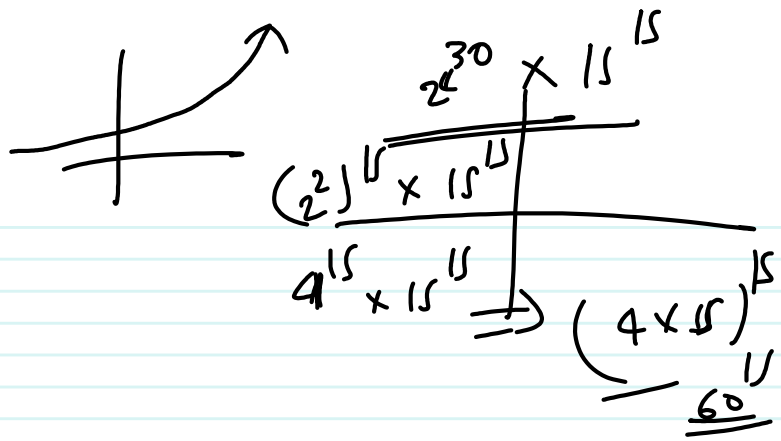
$$(2^3 \times 5^6)^2$$

$$2^6 \times 5^{12}$$

$$\Rightarrow 2^{30} \times 3^{15} \times 5^{15}$$

$$f(x) = e^x$$

$$a > 1, a^x$$



Composition

$$f(x) = 2x, \quad g(n) = x^3 + 5$$

$$(f \circ g)(n) = \underset{\parallel}{f(\underset{\parallel}{g(n)})} = \underset{\parallel}{2 \times g(n)} = 2(n^3 + 5) = 2n^3 + 10$$

f o g

$$f: B \rightarrow A$$

$$g: C \rightarrow \textcircled{B} \rightarrow \text{Range of } g$$

$a \in C \downarrow$

$$f(\underline{x^3 + 5}) = 2(n^3 + 5) = \underline{2n^3 + 10}$$

sho 1 be domain for f.

$$\underline{g(a) = 0}$$

$$\underline{f \circ g} = f(g(n)) \quad f(n) \neq \frac{1}{n}$$

$$f(g(a))$$

$$\underline{g(a) = 0} \quad a \in C$$

$$\underline{f(0) =}$$

$$g(x) = \underline{x-2}$$

$$\underline{f(x) = \frac{1}{x}}$$

$$g(n) = \underline{n-2}$$

$$\textcircled{\underline{f \circ g}} \rightarrow$$

$$\underline{f(g(n))} = f(g(2)) = f(0)$$

$\mathbb{R} \setminus \{2\}$

$$f(n) = \left(\frac{1}{n}\right), \quad \underline{g(n) = \left(\frac{1}{n-2}\right)} \quad n=2$$

$$\underline{(f \circ g)(n)} = f(g(n)) = \frac{1}{g(n)} = \frac{1}{n-2} = \underline{\underline{n-2}}$$

$$\underline{(f \circ g)(n)} = \underline{n-2}$$

$$\underline{\underline{\mathbb{R} \setminus \{2\}}}$$

Inverse of f_n

$$\boxed{e^n \rightarrow \log n}$$

$$\underline{\underline{f: A \rightarrow B}}$$

$$g: B \rightarrow A$$

$$\underline{\underline{(g \circ f)(n) = n}} \quad \wedge \quad \underline{\underline{(f \circ g)(n) = n}}$$

$$\underline{\underline{f(n) = 2n + 5}}$$

$$f: \mathbb{R} \rightarrow \mathbb{R}$$

$$g: \mathbb{R} \rightarrow \mathbb{R} \Rightarrow (f \circ g)(n) = \underline{\underline{n}}$$

$$f(n) = y$$

$$\left. \begin{array}{l} \underline{\underline{f(n) = 2n + 5}} \\ y = 2n + 5 \end{array} \right\}$$

$$\Rightarrow 2n = y - 5$$

$$\Rightarrow \underline{\underline{n = \frac{y-5}{2}}}$$

$$\underline{\underline{g(n) = \frac{n-5}{2}}}$$

$$\underline{\underline{f \circ g = f(g(n)) = 2g(n) + 5}}$$

$$= 2\left(\frac{n-5}{2}\right) + 5 = n - 5 + 5 = \underline{\underline{n}}$$

$$\left. \begin{array}{l} \underline{\underline{f(n) = n^2}} \\ n = 2 \rightarrow 4 \\ n = -2 \rightarrow 4 \end{array} \right\}$$

One-one - In that inverse exist

$$f(n) = \underline{\underline{n^2}}$$

one-one

$$g(n):$$

$$\log_a n$$

$$f(g) =$$